# Generating Cancelable Fingerprint Templates

Nalini K. Ratha, *Fellow*, *IEEE*, Sharat Chikkerur, *Student Member*, *IEEE*, Jonathan H. Connell, *Senior Member*, *IEEE*, and Ruud M. Bolle, *Fellow*, *IEEE*

**Abstract**—Biometrics-based authentication systems offer obvious usability advantages over traditional password and token-based authentication schemes. However, biometrics raises several privacy concerns. A biometric is permanently associated with a user and cannot be changed. Hence, if a biometric identifier is compromised, it is lost forever and possibly for every application where the biometric is used. Moreover, if the same biometric is used in multiple applications, a user can potentially be tracked from one application to the next by cross-matching biometric databases. In this paper, we demonstrate several methods to generate multiple cancelable identifiers from fingerprint images to overcome these problems. In essence, a user can be given as many biometric identifiers as needed by issuing a new transformation "key." The identifiers can be cancelled and replaced when compromised. We empirically compare the performance of several algorithms such as Cartesian, polar, and surface folding transformations of the minutiae positions. It is demonstrated through multiple experiments that we can achieve revocability and prevent cross-matching of biometric databases. It is also shown that the transforms are noninvertible by demonstrating that it is computationally as hard to recover the original biometric identifier from a transformed version as by randomly guessing. Based on these empirical results and a theoretical analysis we conclude that feature-level cancelable biometric construction is practicable in large biometric deployments.

**Index Terms**—Fingerprint identification, image registration, cancelable biometrics, privacy, security.

## 1 Introduction

Securing information and ensuring the privacy of personal identities is a growing concern in today's society. Traditional authentication schemes primarily utilize tokens or depend on some secret knowledge possessed by the user for verifying his or her identity. While these techniques are very popular, they have several limitations. Both token and knowledge-based approaches cannot differentiate between an authorized user and a person having access to the tokens or passwords. In case of knowledge-based authentication systems, managing multiple passwords (i.e., identities) presents usability problems. Biometrics-based authentication schemes using fingerprints, face recognition, etc., overcome these limitations while offering usability advantages and are therefore rapidly extending traditional authentication schemes. However, despite its obvious advantages, the use of biometrics raises several security and privacy concerns as outlined below:

1. Biometrics is authentic but not secret: Unlike passwords and cryptographic keys that are known only to the user, biometrics such as voice, face, signature, and even fingerprints can be easily recorded and potentially misused without the user's consent. There have been several instances where artificial fingerprints [14] have been used to circumvent biometric security systems. Face and voice biometrics are similarly vulnerable to being captured without the user's explicit knowledge. In contrast, tokens and knowledge have to be willingly shared by the user to be compromised.
2. Biometrics cannot be revoked or canceled: Passwords, PINs, etc., can be reset if compromised. Tokens such as credit cards and badges can be replaced if stolen. However, biometrics are permanently associated with the user and cannot be revoked or replaced if compromised. While a user can successively enroll different fingerprints, there is still a limited choice of fingers to choose from. This choice does not exist for other biometric modalities.
3. If a biometric is lost once, it is compromised forever: Biometrics provides usability advantages since it obviates the need to remember and manage multiple passwords/identities. However, this also means that if a biometric is compromised in one application, essentially all applications where the particular biometric is used are compromised.
4. Cross-matching can be used to track individuals without their consent: Since the same biometric might be used for various applications and locations, the user can potentially be tracked if organizations collude and share their respective biometric databases. With traditional authentication schemes, the user can maintain different identities/passwords to prevent this. The fact that a biometric remains the same presents a privacy concern.

Observing that this is an important area of future research, the 2003 US National Science Foundation Workshop on a Biometrics Research Agenda [19] identified "anonymous biometrics" as a privacy enhancing technology of great interest. The techniques that can meet these requirements are also called *cancelable biometrics*. While conceptual frameworks for cancelable biometrics were presented when these

- *N.K. Ratha, J.H. Connell, and R.M. Bolle are with IBM Research, 19 Skyline Drive, Hawthorne, NY 10598. E-mail: {ratha, jconnell, bolle}@us.ibm.com.*
- *S. Chikkerur is with the Department of Electrical Engineering and Computer Science, Massachusetts Institute of Technology, 77 Massachusetts Avenue, Cambridge, MA 02139-4307. E-mail: sharat@mit.edu.*

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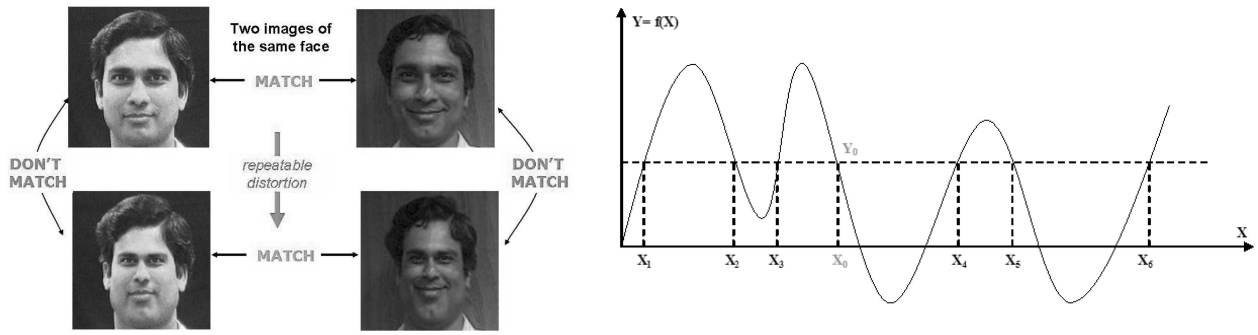


(a) (b)

Fig. 1. (a) Illustration of cancelable biometrics for face recognition. In this case, the face is distorted in the signal domain prior to feature extraction. The distorted version does not match with the original biometric, while the two instances of distorted faces match among themselves. (b) Illustration of feature domain transformation. In this case, each feature (e.g., minutiae position) is transformed using a noninvertible function $Y = f(X)$. For instance, the minutiae position $X_0$ is mapped to $Y_0 = f(X_0)$ as shown. However, if we know $Y_0$, the inverse mapping is a many-to-one transformation. $X_0, X_1, X_2 \ldots X_6$ are all valid inverse mappings to $Y_0$. The complexity of the inverse mapping is exponential in the number of features, making the transform practically noninvertible.

problems were first identified [3], [4], [20], working solutions emerged only recently. A comprehensive review of these techniques can be found in [30].

The topic of this paper, noninvertible (cancelable) transforms, was one of the original solutions to privacy-preserving biometric authentication. Instead of storing the original biometric, it is transformed using a one-way function. The transformed biometric and the transformation are stored either distributed on a smart-card or centrally in a database. The transformation can be performed either in the signal domain (see Fig. 1a) or in the feature domain [20] (Fig. 1b). This construct preserves privacy since it will not be possible (or computationally very hard) to recover the original biometric template using such a transformed version. If a biometric is compromised, it can be simply reenrolled using another transformation function, thus providing revocability. The construct also prevents cross-matching between the databases, since each application using the same biometric uses a different transformation. Another advantage of this approach is that the feature representation is not changed (in both signal and feature domain transformation). This allows the use of existing feature extraction and matching algorithms. Moreover, the approach is backward compatible with legacy biometric authentication installations.

In this paper, we present several noninvertible transforms for constructing multiple identities from a fingerprint template. The basic minutiae features are used in the template. In Section 2, we review the recent literature in this area. The key requirements for constructing a cancelable template and the details of our proposed approach, including accurate registration and various methods of transforming a template, are presented in Section 3. The proposed techniques have been extensively tested on various issues, including cancelability, cross matching, noninvertibility, and loss of accuracy attributed to transforms. We present these results in Section 4 and, in Section 5, we analyze the brute force attack strength. Finally, in Section 6, we summarize our results and conclusions.

## 2 Related Work

After Ratha et al. [20] formally defined the problem of cancelable biometrics (also called revocable biometrics), many alternate solutions have emerged from both the biometric and cryptographic community. We loosely divide the prior work into the following categories:

1. **Biometric salting**. This is similar to password "salting" in conventional crypto-systems. In this approach, before hashing the password $P$ of the user, it is concatenated with a pseudorandom string $S$ and the resulting hash $H(P + S)$ is stored in the database. The addition of the random sequence increases the entropy and, therefore, the security of the password. Biometric salting is based on the same principle. In some instances, the new representation is quantized to derive robust binary cryptographic keys. However, the quantization is practical only because of the additional entropy introduced through the "salt." The defining feature of this category is the addition of user-specific random information to increase the entropy of the biometric template. Further details may be found in [8], [22], [23], [25], [26].
2. **Biometric key generation**. In this approach, a key is derived directly from the biometric signal. The advantage is that there is no need for user-specific keys or tokens as required by "*biometric salting*" methods and that it is therefore scalable. A key $K(B)$, parameterized by the biometric $B$, is stored instead of the actual biometric itself. During verification, it is checked if $K(B') = K(B)$. The major problem with this approach is achieving error tolerance in the key. The defining feature of this category is the attempt to derive robust binary representations (keys) from noisy biometric data without the use of additional information. Further details may be found in [6], [15], [16], [31].
3. **Fuzzy schemes**. Another approach for constructing cancelable templates involves the use of public auxiliary information $P$ (also called helper data, shielding functions, or fuzzy extractors), which is combined with biometric information to reduce the intrauser variation. The following is the defining feature of this category: The schemes define a metric $d(B, B')$ (e.g., Hamming, Euclidian, set distance, etc.) on noisy biometric data $B$ and $B'$. Further, the schemes use generating and reproducing functions,

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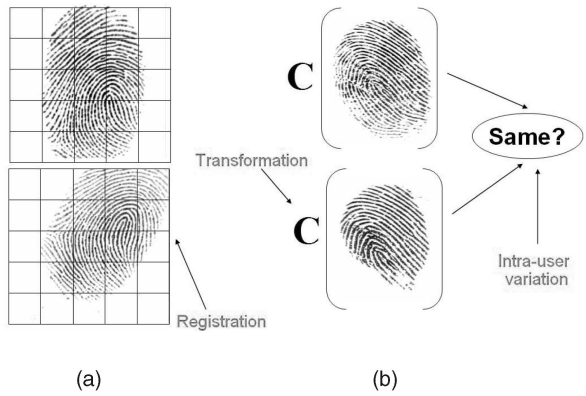


Fig. 2. Challenges for implementing feature level distortion. (a) Registration is required for repeatability. (b) Intrauser variation must be accomodated.

call them $Gen$ and $Rep$. Broadly speaking, the generating function takes the biometric data along with user specific key/information $K$ to produce a public string $P$ and a secret string $S$, $Gen(B, K) \rightarrow < S, P >$. The reproducing function $Rep$ takes the public string along with another biometric measurement to reproduce the secret string $Rep(B', P) \rightarrow S$. In other words, fuzzy schemes extract some randomness $S$ from $B$ and then successfully reproduce $S$ as long as $d(B, B') \leq \epsilon$. Further details may be found in [7], [10], [11], [13], [28], [29].

4. **Noninvertible transforms**. In this technique, instead of storing the original biometric, the biometric is transformed using a one-way function. The transformation occurs in the same signal or feature space as the original biometric. Our proposed approach [9] falls under this category. The defining feature of this category is its backward compatibility with existing matching algorithms and legacy installations. Further details may be found in [1], [24].

## 3 Cancelable Templates for Fingerprints

We describe several techniques for designing cancelable templates for fingerprint biometrics using one-way transformations in the feature domain. Instead of storing the original minutiae features, the minutiae location and orientations that are stored are transformed irreversibly.

In the following, let $\mathcal{M}$ be a fingerprint matching algorithm, which, in this paper, is assumed to be a tolerance box matcher like [21]. The matcher acting on two biometrics $x_1$ and $x_2$ gives the matching or similarity score $0 \leq \mathcal{M}(x_1, x_2) \leq 1$. As shown in Fig. 2, there are several challenges to be overcome before successfully designing a cancelable transform $C$ that transforms biometric $x$ into a cancelable biometric $C(x)$. Similar to [1], we formalize the challenges in terms of requirements on $\mathcal{M}(x_1, x_2)$:

1. **Registration**. For the cancelable transform $C$ to be repeatable from one instance $y_1$ of a biometric to the next instance $y_2$ of the same biometric, the biometric signals $y_1$ and $y_2$ must be positioned in the same coordinate system each time. For fingerprints, this means that both $y_1$ and $y_2$ have to be first rotated and translated by coordinate transform $T_1$ and $T_2$, respectively, such that the signals $x_1 = T_1(y_1)$ and $x_2 = T_2(y_2)$ overlap. Hence, it is required that, prior to the cancelable transformation $C$, the impressions $y_1$ and $y_2$ are registered such that corresponding minutiae match as well as possible.
2. **Intrauser variability tolerance**. Another problem to contend with, even after registration, is the intrauser variability that is present in biometric signals. The features obtained after the transformation should be robust with regard to to this variation, in that the probability of a false reject should ideally not increase in the transformed domain, i.e., when $x_1$ and $x_2$ match

$$\mathcal{M}(x_1, x_2) > t \Rightarrow \mathcal{M}(C(x_1), C(x_2)) > t. \quad (1)$$

3. **Entropy retention**. The transformed version should not lose any individuality. The intrinsic strength of a transformed biometric $C(x)$ should be comparable to the original biometric $x$, that is, the probability of a false accept should not increase in the transformed domain. Ideally, we should have that, if $x_1$ and $x_2$ do not match,

$$\mathcal{M}(x_1, x_2) \leq t \Rightarrow \mathcal{M}(C(x_1), C(x_2)) \leq t, \quad (2)$$

where $t$ is the decision threshold. In [1], the combination of (1) and (2) is called $\epsilon$-match preserving.
4. **Transformation function design**. The transform $C$ has to further satisfy the following conditions:

   a. The transformed version $C(x)$ of the biometric $x$ should not match the original, i.e.,

   $$\mathcal{M}(C(x), x) \leq t, \quad (3)$$

   where, again, $t$ is the decision threshold for matching algorithm $\mathcal{M}$.
   b. Multiple personalities $C_1(x)$ and $C_2(x)$ generated from the same template $x$ of the same biometric should not match

   $$\mathcal{M}(C_1(x), C_2(x)) \leq t. \quad (4)$$

   This property of the transform prevents cross-matching between databases when different cancelable transforms $C_1$ and $C_2$ are used. This is related to the two transformations $C_1$ and $C_2$ being $\kappa$-distinct in [1].
   c. The original biometric or template $x$ should not be recoverable from the transformed one $C(x)$ in that the inverse transform $C^{-1}$ should not exist. This property preserves the privacy of the original template $x$ since it is not stored.

### 3.1 Registration Prior to Transformation

The first important step in the application of a cancelable transform is the process of registering the image. For the transform to be repeatable, the minutiae positions have to be measured with regard to the same coordinate system. This can be accomplished by estimating the position and orientation of the singular points (core and delta) and expressing the minutiae positions and angles with respect to these points. There have been several approaches to determine the core and delta [2], [12], [17], [27], but precise estimation remains a difficult problem.

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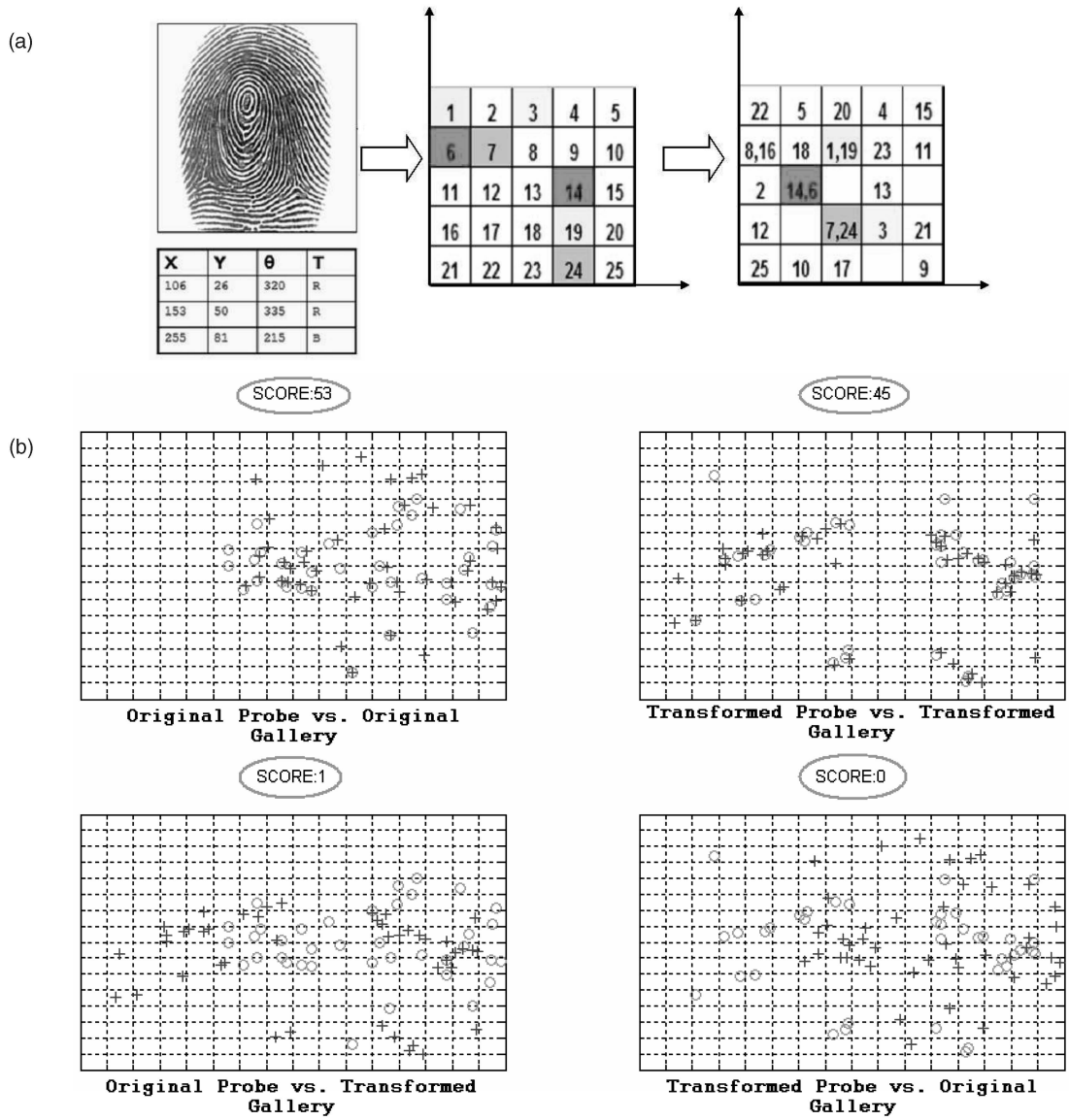


Fig. 3. In a Cartesian transformation, the space is divided into cells of equal size. (a) The transformation maps each cell to some random cell with collisions. (b) In this example of Cartesian transformation, it can be seen that the transformed version does not match with the original (lower right).

We use an algorithm, which is motivated by Nilsson [18]. He proposed a new approach which relies on detecting the parabolic and triangular symmetry associated with core and delta points. Filtering is done on complex images associated with the orientation tensor [17]. The algorithm finds the most likely singular point positions and the associated directions. We provide several improvements of this approach [5].

Once the global registration has been established using the singular point location, the minutiae feature points can be transformed consistently across multiple instances. While the general idea of cancelable biometrics is to irreversibly transform the minutiae feature positions and orientations, the transform itself can be implemented in several different ways. We propose three transform methods: Cartesian, polar, and surface folding.

### 3.2 Cartesian Transformation

In the Cartesian transformation, the minutiae positions are measured in rectangular coordinates with reference to the position of the singular point. The $x$-axis is aligned with the orientation of the singular point. This coordinate system is divided into *cells* of fixed size (with a total of $H \times W$ cells as illustrated in Fig. 3). The cells are numbered in a fixed sequence. The process of transformation consists of changing the cell positions (see Fig. 3). Additionally, the cells may be rotated in multiples of 90 degrees after transposition. However, in this paper, only position displacement is considered. When the cell positions are changed, all the minutiae within the cells retain their relative positions.

The transformation is not a strict permutation since the condition of irreversibility requires that more than one cell be mapped to the same cell. In our case, the cell mapping is governed by a mapping matrix $M$. The positions of the cells, before $C$ and after $C'$ transformation, can be written simply as

$$C' = CM. \tag{5}$$

For instance, consider a simplified process where the coordinates are divided into just four cells in a $2 \times 2$ pattern. The transformation of the matrix $M$ is

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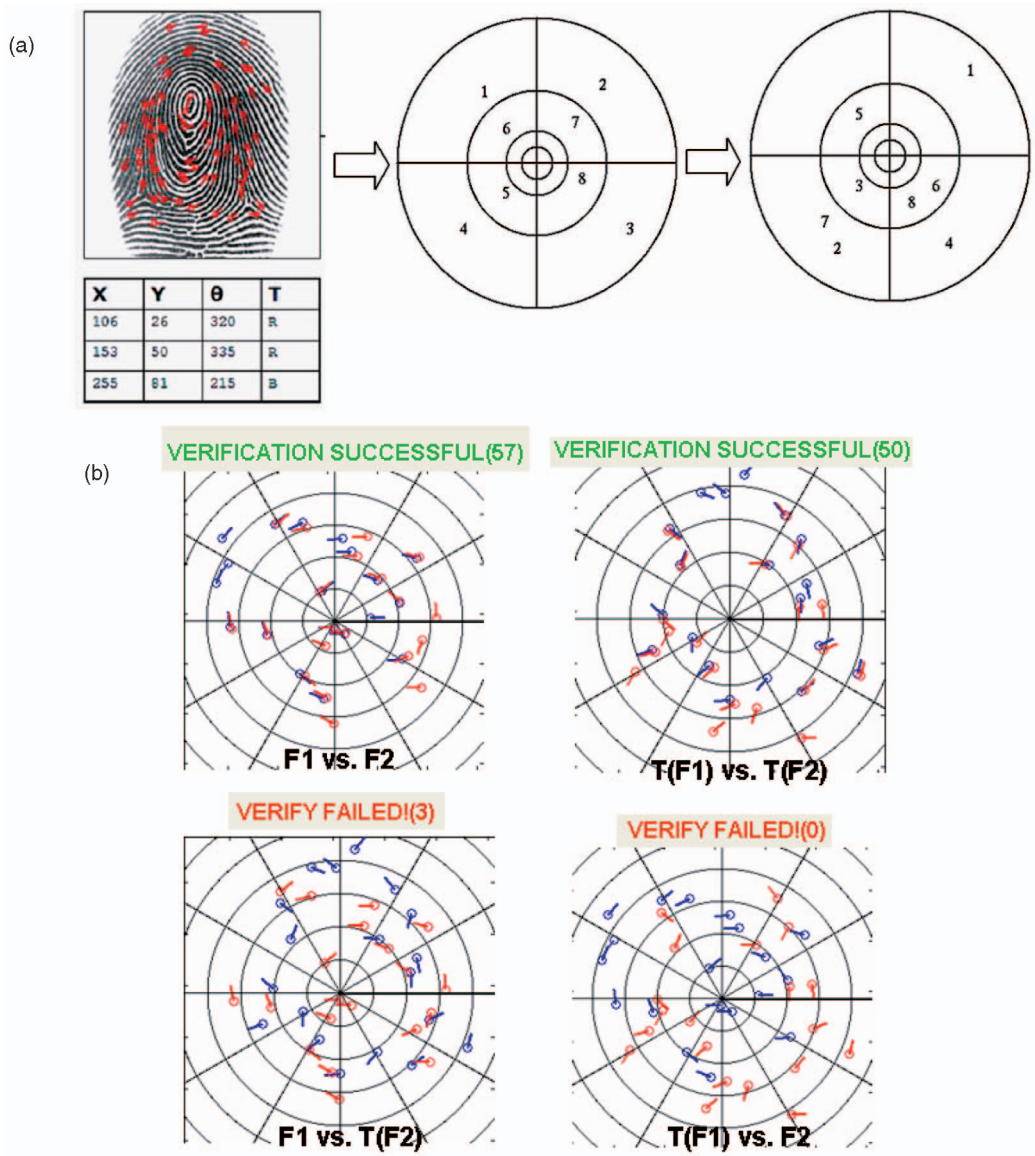


Fig. 4. (a) For the polar transform, the feature space is divided into sectors. Each sector is mapped into some other random sector after transformation. (b) In this example of polar transformation, it can be seen that the transformation and the original do not match, while two similarly transformed instances match well (upper right).

$$\begin{bmatrix} 1 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 2 & 4 & 3 \end{bmatrix}. \quad (6)$$

It can be seen that both cells 1 and 4 are mapped to cell 3, so that, even if both the transformation and the transformed pattern are known, it is impossible to determine which minutiae in cell 3 are from the original cell 1 or from the original cell 4.

### 3.3 Polar Transformation

In this method, the minutiae positions are measured in polar coordinates with reference to the core position. The angles are measured with reference to the orientation of the core. The coordinate space is now divided into polar *sectors* ($L$ levels and $S$ angles) that are numbered in a sequence (see Fig. 4). The process of transformation now consists of changing the sector positions. The minutiae angles also change in accordance to the difference in the sector positions before and after transformation.

Unlike the Cartesian transformation, unconstrained mapping is not feasible in polar coordinates since the angular shift $d\theta$ is transformed to a positional shift $r\,d\theta$ at a distance $r$ from the core. This leads to a situation where minutiae pairs that occur within a tolerance distance of each other before transformation no longer match after transformation due to the large divergence that occurs away from the core. Therefore, in the polar transformation, the mapping is governed by a translation key $(1 \times LS)$ that defines the cell transformation. The positions of the sectors before and after transformation are related as

$$C' = C + M. \quad (7)$$

Here, consecutively numbered sectors are generally in close proximity and the elements of the transform matrix are constrained to have low absolute values. In particular, the

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translation is constrained to prevent performance (error rate) overhead.

For instance, consider a simplified transformation where $L = 2$ and $S = 2$. The transformation through the matrix $M$ is

$$\begin{bmatrix} 1 & 2 & 3 & 4 \end{bmatrix} + \begin{bmatrix} +3 & -1 & +1 & -2 \end{bmatrix} = \begin{bmatrix} 4 & 1 & 4 & 2 \end{bmatrix}. \tag{8}$$

This mapping is many-to-one. In the example, after transformation, both original sectors 1 and 3 are mapped to sector 4. Thus, even given the transformation and the transformed pattern, it is impossible to invert the transformed pattern.

A similar transformation is also applied to the minutiae angles. Unlike the Cartesian transformation, to a large extent, this does not alter the natural distribution of minutiae points (see Fig. 4).

### 3.4 Functional Transformation

The primary shortcoming of both the Cartesian and polar transformation is that a small change in minutiae position in the original fingerprint can lead to a large change in minutiae position after transformation if the point crosses a sharp boundary. This renders the matching process vulnerable to increased intrauser variation. It seems that a smooth but noninvertible transformation would achieve a higher performance. In addition, this smooth transformation should be largely unconstrained and a many-to-one mapping to satisfy the noninvertibility condition. In this section, we explore a family of functions giving a locally smooth transformation of the minutiae positions. The transformation has a parametric form that is governed by a random key. Like before, the transformation is applied after aligning the input fingerprint using its core location and orientation. The requirement of cancelability puts several constraints on the parametric function:

1. The transformation should be locally smooth to ensure that a small change in a minutiae position before transformation leads to a small change in the minutiae position after transformation.
2. The transformation should not be globally smooth. If the minutiae positions after transformation are highly correlated to minutiae positions before transformation, the transformation can be inverted easily. Moreover, the transformation should be many-to-one to make sure it cannot be uniquely inverted to recover the original minutiae pattern.
3. Since minutiae-based matchers tolerate a certain amount of uncertainty in the minutiae position and orientation to account for feature extraction inaccuracies, each minutiae position must be pushed outside the tolerance limit of the matcher after transformation. Thus, we need to ensure that there is a minimum amount of translation during the transformation.

Using such "locally smooth but globally not smooth" functions

$$X' = x + f_X(x, y), \tag{9}$$

$$Y' = y + g_Y(x, y), \tag{10}$$

$$\Theta' = \mathrm{mod}(\Theta + h_\Theta(x, y), 2\pi), \tag{11}$$

the position and orientation of the minutiae are altered. Note that all transformations are functions of both $x$ and $y$, in contrast to [20], which originally proposed separable transformation functions.

#### 3.4.1 Function Design

It is not obvious how to design these functions to satisfy all the constraints. The requirement of a minimum translation poses a severe restriction on the choice of the function. Most spatial functions cannot achieve this unless we add a fixed offset at all locations. However, this does not satisfy the "globally not smooth" constraint imposed on the function. On the other hand, if we scale the function such that the minimum value of the function has a given offset, then it is very likely that the function will not be locally smooth since the scaling magnifies the function gradient at all points. Furthermore, to satisfy the noninvertibility condition, the function should have *folding* or transformed regions that originate from multiple locations in the original space. In this section, we will outline several alternatives for "locally smooth but not globally smooth" functions as in Fig. 5.

We solve the problem of folding and minimum translation by modeling the direction of translation instead of its magnitude. We then ensure that the minutiae move in this direction by at least a fixed distance. The direction field is chosen to be locally smooth but globally not smooth. More rigorously, we model the translation using a vector valued function $\vec{F}(x, y)$ whose phase determines the direction of translation. The extent of translation is given by the magnitude $|\vec{F}|$ or, alternatively, another vector valued function $\vec{G}(x, y)$. Some examples of how to design such a function are as follows:

1. The vector function $\vec{F}$ is an electric potential field (see Fig. 6a) parameterized by a random distribution of charges. The magnitude and phase of this vector function are given by

$$|\vec{F}| = \left| \sum_{i=1}^{K} \frac{q_i(z - z_i)}{|(z - z_i)|^3} \right|, \tag{12}$$

$$\Phi(x, y) = \frac{1}{2} \arg\left\{ \sum_{i=1}^{K} \frac{q_i(z - z_i)}{|(z - z_i)|^3} \right\}. \tag{13}$$

   Here, $z = x + iy$ is the position vector. The random key $K = z_1, z_2, \ldots, z_K, q_1, q_2, \ldots, q_K$ determines the position and magnitude of the charges.
2. As an alternate formulation, we represent the phase (direction of translation) as the gradient of a mixture of Gaussian kernels and the magnitude (extent of translation) as the scaled value of the mixture. In fact, we can use separate mixtures for evaluating the phase and magnitude to avoid any form of correlation between them (see Fig. 6b)

$$|\vec{F}(z)| = \sum_{i=1}^{K} \frac{\pi_i}{|2\pi\Lambda_i|} \exp\left\{ -\frac{1}{2}(z - \mu_i)^T \Lambda_i^{-1} (z - \mu_i)) \right\}, \tag{14}$$

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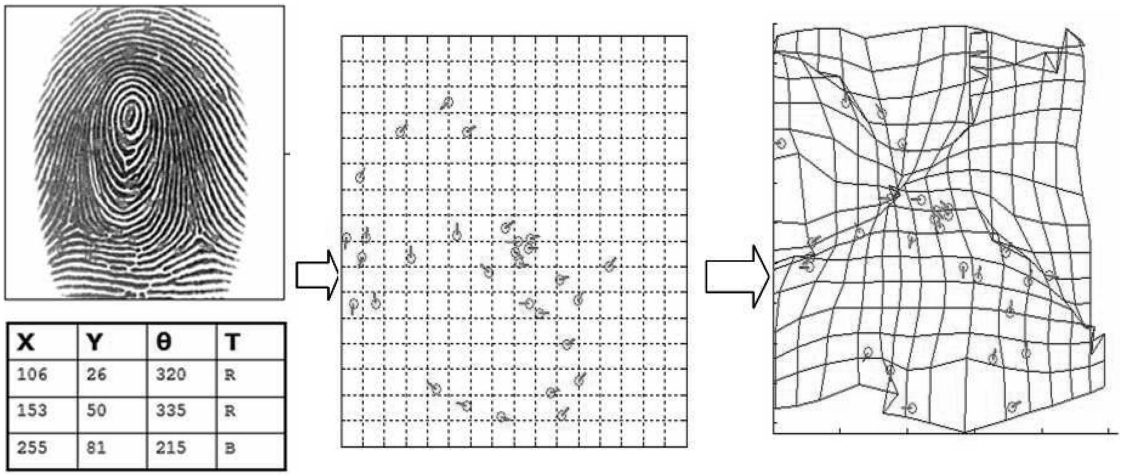


Fig. 5. In a surface folding transformation, both the position and the orientation of the minutiae are changed by some parametric transfer function. Conceptually, the minutiae are embedded in a sheet which is then crumpled. This function is locally smooth but globally not smooth.

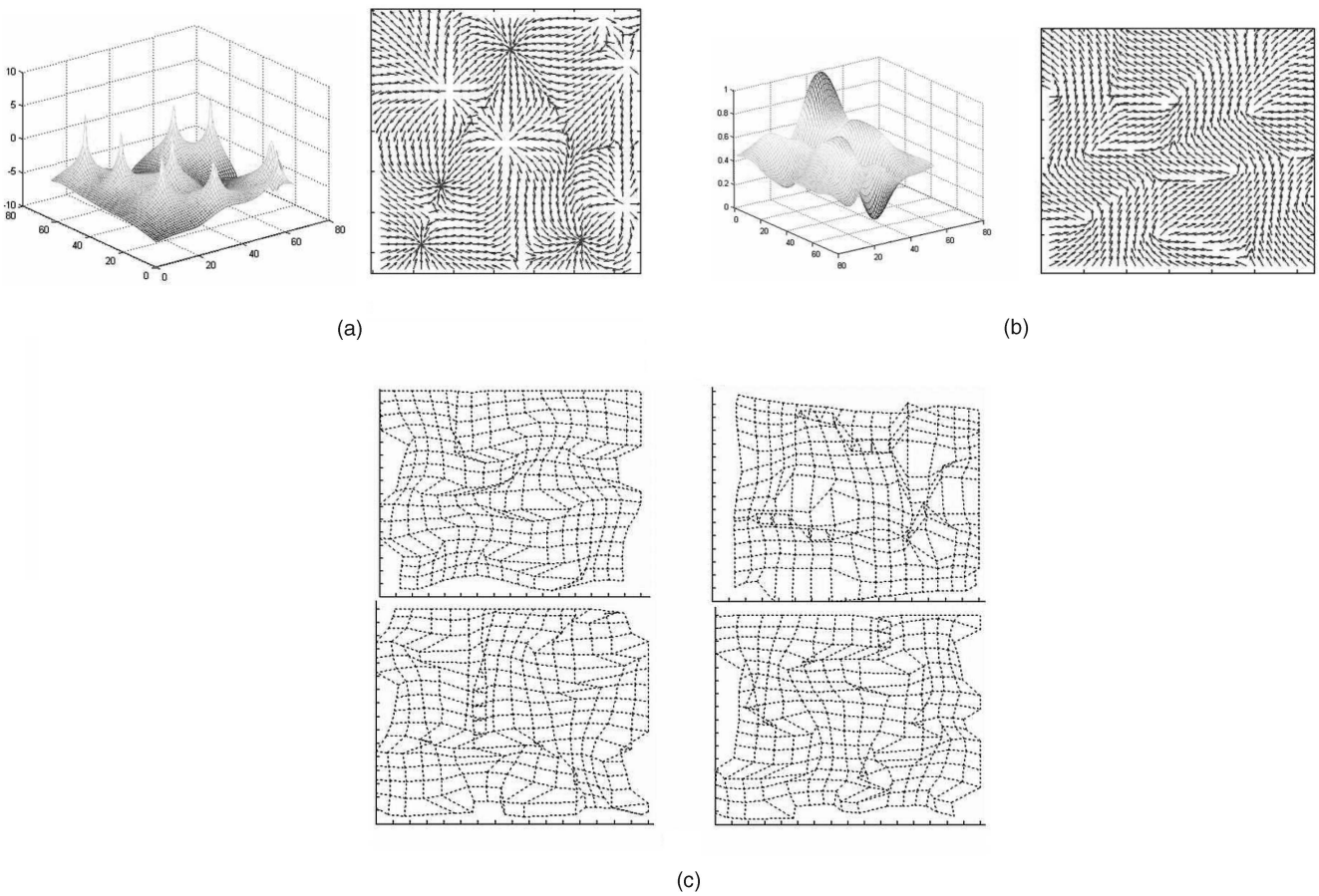


Fig. 6. There are several ways to obtain the transformation function. (a) The function is parameterized by a random charge distribution. The phase angle of the resulting vector decides the direction of movement and the magnitude parameterizes the extent of movement. (b) The function is parameterized by a random mixture of Gaussian kernels. The direction of movement is parameterized by the gradient vector of this surface. (c) Examples of some warping functions generated by the Gaussian method. Only the coordinate transformation is shown here.

$$\Phi_F(z) = \frac{1}{2}\arg\left\{\nabla\vec{F}\right\} + \Phi_{rand}. \tag{15}$$

Here, $z$ represents the position vector $[x, y]^T$. The key defines the parameters of the distributions such as the weights $\pi_i$, covariances $\Lambda_i$, the center of the kernels $\mu_i$, and the random phase offset $\Phi_{rand}$. The transformation is given by

$$X' = x + K|\vec{G}(x,y)| + K\cos(\Phi_F(x,y)), \tag{16}$$

$$Y' = y + K|\vec{G}(x,y)| + K\sin(\Phi_F(x,y)), \tag{17}$$

$$\Theta' = \mathrm{mod}(\Theta + \Phi_G(x,y) + \Phi_{rand}, 2\pi). \tag{18}$$

Hence, for the displacement, the magnitude function $|\vec{G}|$ and the phase function $\Phi_F$ of $\vec{F}$ are used; for the angle, the unused phase function $\Phi_G$ of $\vec{G}$ is used.

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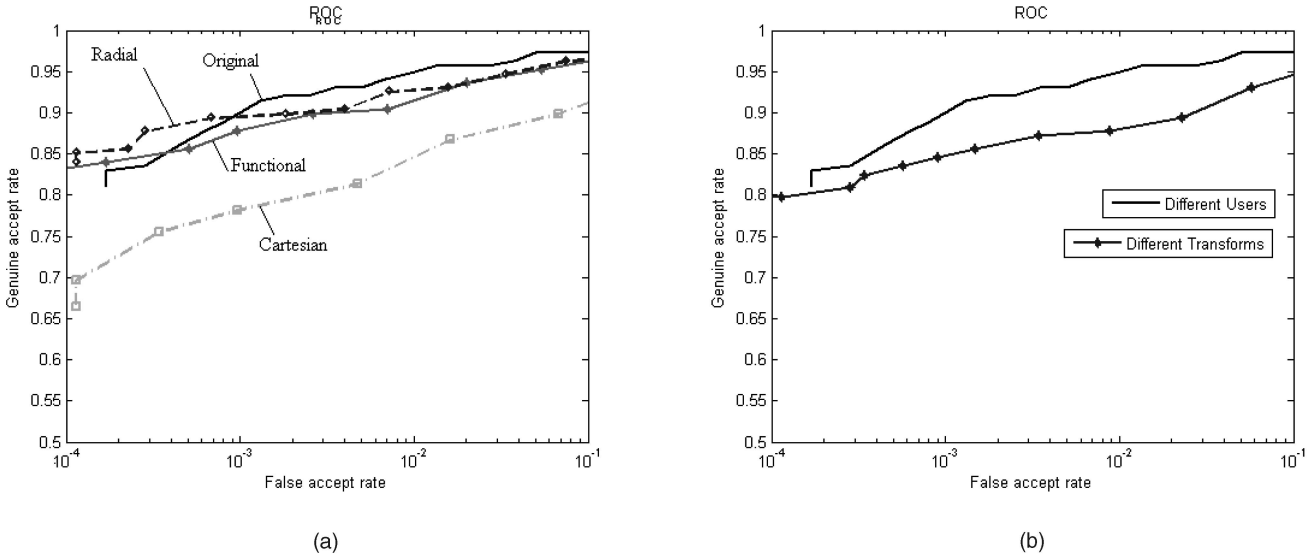


Fig. 7. Cancelable transform performance: (a) Comparison of the matching performance without transformation and with three different classes of cancelable transform. These performance curves were generated using 188 pairs of fingerprints and a tolerance-box matcher. (b) Distinctiveness of transforms. The solid curve shows the performance obtained with the original untransformed database. The second curve shows a pseudo-ROC where the prints were matched solely against various different functional transforms of themselves. It can be seen that this does not substantially impact performance, suggesting that the original prints and their transforms are uncorrelated.

In practice, we typically use 24 Gaussians all with the same isotropic standard deviation of 50 pixels. The centers of the Gaussians are placed randomly (flat distribution) in the $512 \times 512$ image space and each Gaussian is given a peak magnitude of either $+1$ or $-1$. Therefore, the key for such a surface consists of 456 bits $(24 \times (9 + 9 + 1))$. The additive superposition of all the functions is taken to generate the surface. For cancelable fingerprints, we typically generate two such surfaces. One is used to pick the direction in which to move each minutiae by finding the orientation of the local gradient. The other surface $S(x, y)$ is used to select how far to move the minutiae point. For this, the surface values are shifted and scaled so that the highest peak has a height of 1 and the lowest valley has a height of 0. The translation is then calculated as a minimum pixel distance plus a scaled version of the surface value at the minutiae location, typically $30 + 30 \times S(x, y)$.

## 4 Experimental Study

We performed an empirical study using the IBM-99 optical database for transform analysis. This consists of 188 fingerprint pairs, after rejecting poor quality fingerprints.

We are concerned with several questions about the cancelable transforms:

1. What impact does the transformation have on the matching performance?
2. How do the various transformation methods compare with respect to performance?
3. How sensitive is the performance to the choice of transform parameter keys?
4. How distinctive are the transforms? (Does one transform match against other persons or against other transforms?)
5. What is the approximate brute force strength against an invertibility attack?

In this section, we present empirical results for most of these questions. We theoretically derive the approximate strength to answer the last question (see Section 5).

Of course, the overall cancelable matching characteristics also depend on the quality of the registration process and the parameters of the transform itself.

### 4.1 Performance Impact

To measure the effect that cancelable transforms have on the overall fingerprint matching performance, we use a tolerance box matcher similar to the one described in [21]. The only matcher parameter that influences the effectiveness of the cancelable transforms is the tolerance box size (bounding distance/tolerance limit), i.e., the maximum distance between two minutiae considered to be a match. In principle, the effectiveness of the transforms is not affected by any other parameter of the particular fingerprint matcher.

For the 188 mated pairs, we generate the ROC curve for raw (untransformed) prints and for each of the three transform methods. For each individual, we assign a different transform from the given class (i.e., 188 different transforms). This reflects how the system would be used in the real world. Suppose the same transform were assigned to all prints. Then, if a hacker obtains a fraction of the database, there are many examples from which to attempt to invert this one transform. If, instead, there are as many transforms as individuals, the amount of available data for cracking each transform is decreased significantly. Moreover, even in the case that everyone starts out with the same transform, over time, certain individuals would need to cancel their original enrollment and obtain a replacement transform. Thus, in the long run, most people will end up with different transforms anyhow.

Fig. 7a illustrates the effect that the various transform methods have on the overall accuracy of the matcher system. The results show that the performance decreases only slightly, and also indicate that the surface folding transform is preferred. It performs noticeably better than the Cartesian version and is comparable to the polar

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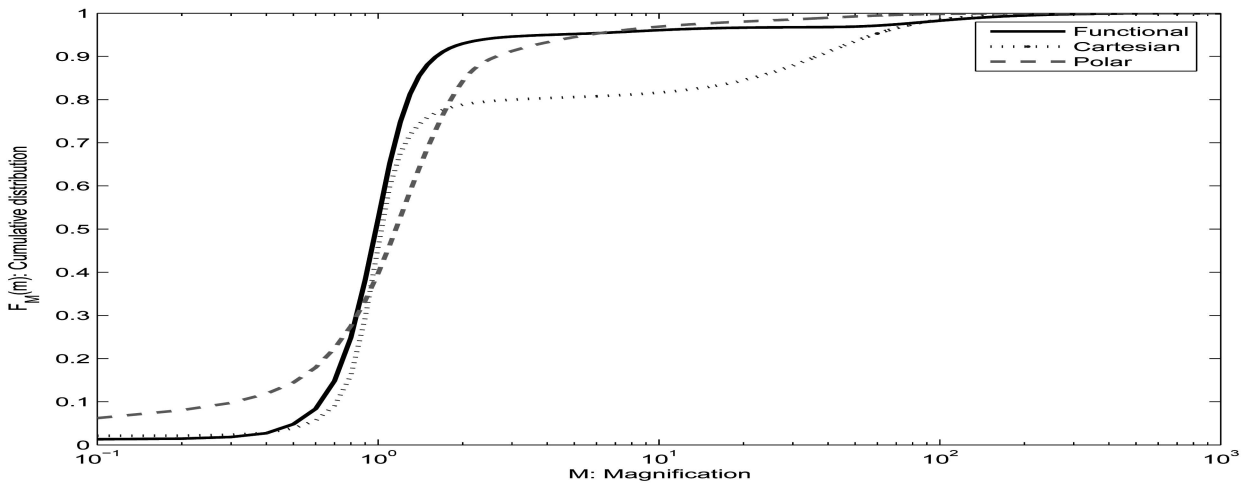


Fig. 8. This cumulative histogram shows how the variation in the position of a minutiae affects the variation of its position in the transformed image. The $x$-axis records the amount by which the perturbation is magnified (or possibly reduced) in the output.

version. However, there are far more possible transforms for the surface folding method as opposed to the tightly constrained polar method.

The relatively poor performance of the Cartesian transform can be explained by the additional intrauser variability introduced due to cell swapping. Since the transformation function is not locally smooth, if a minutiae close to the border of a cell crosses over between instances (due to intrauser variation), it will not match with the corresponding minutiae after transformation. The probability of cell exchanges is proportional to the number of minutiae that fall near the cell boundaries. If we assume that the minutiae can shift $\Delta$ pixels in either direction, a fraction $1 - \frac{(X-2\Delta)^2}{X^2}$ of the area corresponds to border regions where cell exchanges can occur. In our case, the cell width is chosen to be $X = 32$. If we chose $\Delta = 4$, the cell exchange probability will be close to 42 percent. With $\Delta = 8$, the cell exchange probability will be close to 75 percent! Due to the nonlinear and discontinuous nature of the transformation, small changes in the minutiae location before transformation may result in very large changes in position after transformation, causing a loss of accuracy.

### 4.2 Distinctiveness

Another question we wish to answer is whether the original fingerprint and its transformed version are correlated. To test this, we selected one mated pair and enrolled this pair plus 187 Gaussian transformed versions of the pair in the database (i.e., we selected 187 transforms and applied the same transform to each member of the pair). This gives a gallery size equivalent to the original experiment (187 other pairs = other prints plus the two that should match). Using just the two original prints in the pair as probes, we computed the matching statistics. The results are shown in Fig. 7b along with the ROC for the original, untransformed database. As can be seen, the performance is nearly the same. Thus, the transformed prints are no more likely to match the original prints than completely different individuals are. This supports our contention that changing transforms essentially issues the user a new fingerprint.

### 4.3 Local Smoothness

We discussed how the sharp boundaries in the Cartesian and polar schemes can sometimes move a minutiae point to a totally different part of the image if its detected position changes just slightly. The surface folding transform was designed to eliminate this effect and we would like to verify that it does indeed. To test this, we perturbed each minutiae by a vector governed by a normal distribution with a standard deviation of four pixels. This is roughly equivalent to the variation seen in detected positions between two impressions of the same finger. We then transformed both the original print and its perturbed version and compared the distances between corresponding transformed minutiae.

In particular, we computed a "magnification factor" based on the distance between corresponding minutiae after transformation versus before transformation. Suppose a particular minutiae point in the perturbed image is displaced by seven pixels from its original position in the input image. After transforming both images, the distance between the transformed points is found to be 12 pixels. In this case, a magnification factor of 1.7 will be reported. These factors can be histogrammed to characterize the local smoothness of the transform method. For a null transform, we would expect a spike at 1, signalling that the output shift is exactly the same as the input shift. For the Cartesian transform, we would also expect to see a spike at 1—within a block, the perturbation of a minutiae will be preserved—and a number of other spikes at much higher magnifications. These arise from a minutiae being shifted across a block boundary and ending up very far from its source.

Fig. 8 shows the histogram and the cumulative distribution of the magnification factor computed over all the 188 pairs and 188 different transforms of the original experiment in Fig. 7a. We see that the peak of the distribution is at unity as expected. The Cartesian transform has nearly 20 percent of the perturbation magnified by a factor greater than 1 as a consequence of the cell exchanges. The polar transform has a small percentage of minutiae with magnification factor of less than 1. This is caused by minutiae moving from sectors with larger radii to sectors with smaller radii. The functional/surface-folding transform exhibits close to ideal behavior, with a peak around unity and more than 95 percent of the minutiae around this value. This, in turn, has ramifications for our tolerance-box matcher and suggests that the acceptance

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TABLE 1
To Measure the Degree of Surface Deformation, We Determined How Much of the Local Image Structure Was Preserved

| neighbors changed | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| percentage of points | 91.7% | 0.66% | 2.36% | 2.04 % | 3.24% |

*For each minutiae point, we counted how many of its nearest neighbors changed after transformation.*

boxes need to be made only slightly larger to accommodate this noise magnification. Note that, for the experiments of Fig. 7, no such adjustment was made. (We could use a similar analysis to help us design transforms by looking at, say, how this noise sensitivity varies with the number of Gaussians or with the surface scaling factor.)

### 4.4 Noninvertibility

We want to make it hard to invert the transformed version of the points back into the original point configuration. To be strictly noninvertible it must be impossible to create a function that takes a transformed point and regenerates a unique input point. For the surface folding transform, this can occur in two ways: First, if a region of space (say, a square of $50 \times 50$ pixels) is mapped to a smaller region of space (say, a square of $20 \times 20$ pixels), this "region shrinking" can introduce an ambiguity between points. Two nearby points in the input image may map to exactly the same (quantized) point in the output image. However, minutiae points are rarely closer than one ridge spacing, so the impact of this phenomenon is minimal. More interesting is the case where the warping surface folds back over itself. This can be seen in the upper right-hand portion of the warp field illustrated in Fig. 5. In such cases, two relatively large regions of the input image overlap in the output image. Thus, if one selects some point in the transformed image that lies in such a region, it is impossible to tell which of the two original disjoint input regions the point belongs to. Unlike shrinking, this phenomenon does occur with minutiae points and is a prime source of "strong" noninvertibility.

To measure the degree of such "folding" in the transform, we compute the four nearest neighbors of each minutiae point before and after transformation. Using this local information, a complete mesh of the minutiae points can be constructed. If the meshes before and after the transformation have the same connectivity, inverting the transform is simply a matter of "flattening" the sheet (perhaps using elastic graph matching). However, if the meshes are different, our intuition is that determining the proper point correspondences and, hence, inverting the transform, is much harder.

Table 1 shows the results when each of the 376 prints is distorted with a different surface folding transform (similar to Fig. 7b). Here, we have aggregated the results from all the minutiae in all the prints (about 10,000 minutiae altogether). For each minutiae, we checked how many of its original neighbors were altered after transformation. As can be seen, the local structure of the image is largely preserved, but some fraction of minutiae (8 percent) do have their neighborhoods perturbed. Ideally, we would like to see a somewhat higher percentages of the minutiae having at least one neighbor changed since these indirectly signal the edges of "folds" in the warp field induced by the transform. Perhaps increasing the minimum move distance (currently 30 pixels) would lead to increased folding and, hence, stronger noninvertibility.

## 5 Strength Analysis

A theoretical analysis of the brute force attack strength of the three transformations is presented in this section.

### 5.1 Cartesian Transformation

The binary representation of the exchange matrix is convenient from a storage perspective. It also gives us a first order approximation of the information embedded in the key. Each column of the matrix encodes $\log_2(HW)$ bits. Thus, the total information content has an upper bound of $HW \log_2(HW)$ bits. This is a very loose upper bound since, in reality, we cannot expect all cells to be occupied. If we consider the approximate strength of the transformation process, each resulting cell after the transformation could have originated from $HW$ possible source cells. Therefore, a brute force attack would have to try roughly $HW^{(HW)}$ possibilities (corresponding to $HW \log_2(HW)$ bits).

### 5.2 Polar Transformation

The ROC shows that the polar transformation has lower degradation in performance compared to Cartesian transformation. Thus, in analyzing this transform, our primary concern is to ascertain that the transformed template is not correlated to the original template in any fashion so that inverting the transformation is easier than a blind brute force attack. If this is true, the transform is not a very useful one.

For a first-order approximation, we can associate each minutiae with $\log_2(Kd)$ bits of information [20], where $K$ is the number of discretized minutiae positions and $d$ is the number of unique directions. For a brute force attack to be successful, the attacker has to match only $m$ of the $N$ minutiae present in the reference print, so that the matching score $s = \frac{m^2}{mN}$ exceeds the threshold. Let us assume that an average fingerprint has around 35-40 minutiae and that $\log_2(Kd) \approx 10$ bits. If we also pessimistically assume that we need to cross a threshold of 0.25, then we find that the brute-force attack has a complexity of 70-80 bits.

But, this is assuming that the minutiae are uniformly distributed in the fingerprint. This over-simplified assumption is not empirically validated. After examining the statistics of the normalized (after registration) minutiae distribution, our empirical observations (cf. Fig. 14.9 in [4]) have shown that more than half of the minutiae cluster at a distance of 50-150 pixels from the core, whereas the minutiae angle has a peak around 0 degree. Thus, by selectively attacking the minutiae in this cluster (corresponding to 36 sectors and eight orientations in our implementation), we are guaranteed to match at least half the reference minutiae. The strength of a brute-force attack is thus reduced to $8 \log_2(36 \times 8) \approx 64$ bits. So, it is clearly seen that, while the polar transformation performs better than the Cartesian transformation in terms of error rates, the strength of the fingerprint is severely compromised.

### 5.3 Surface Folding Transform

Since a functional transformation not does incur a large loss in accuracy unlike the other transforms, we are primarily concerned with the security strength (even approximate) of such a transformation. While we have demonstated such security using empirical analysis (Fig. 7b), we also attempt a first-order theoretical approximation. For our approximate

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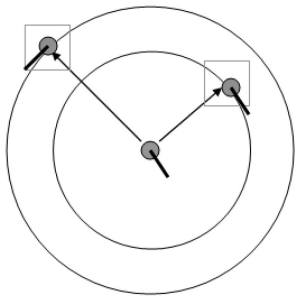

Fig. 9. An approximate model for inverting a point transformed by local surface folding.

analysis, we make the following simplifying assumptions about the transformation:

1. The function can be approximated as a random offset in a random direction determined by the parameterized vector function $\vec{F}(x, y)$. Thus, for analysis, we use the much simpler transformation function

$$X' = x + (K + X_{rand}) \cos(\Phi(x, y)), \tag{19}$$

$$Y' = y + (K + Y_{rand}) \sin(\Phi(x, y)), \tag{20}$$

$$\Theta' = \mathrm{mod}(\Theta + h_\Theta(x, y) + \Phi_{rand}, 2\pi). \tag{21}$$

   This simplification is justified since it provides us with a lower bound on the invertibility strength. By using the transformation of (16), the invertibility will only be harder.
2. We assume that the matcher cannot distinguish minutiae that are within $\delta r$ pixels and oriented $\delta\theta$ of each other.
3. The adversary has to attack only $m$ out of the $N$ minutiae stored to break the transform (corresponding to exceeding the score threshold).

With reference to Fig. 9, the minutiae at the center could have originated from any position in the polar band shown. If we discretize the possibilities using $\delta r = 10$, $\delta\theta = 30$, and $X_{rand} = Y_{rand} = \pm 15$, we find that each transformed minutiae encodes about eight bits of information. For a successful attack, the adversary has to overcome

$$8m - \log_2 \left\{ \binom{N}{m} \right\} \text{ bits} \tag{22}$$

of possibilities. With $m = 12$, $N = 35$, we get an approximate invertibilty strength of 66 bits. Although a gross simplification, this is much harder than a straight brute force attack, thus indicating that the transform does not provide any information about the original template.

## 6 Conclusions and Future Work

While biometrics presents obvious advantages over password and token-based security, the security and privacy concerns that biometric authentication raises need to be addressed. We outlined several advances that originated both from the cryptographic and biometric community to address this problem. In particular, we outlined the advantages of cancelable biometrics over other approaches and presented a case study of applying this technique to a fingerprint database. We studied several alternatives, such as Cartesian, polar, and functional transformation, and compared their relative merits empirically. We also presented a brief and approximate analysis of their strengths. Our experimental analysis shows that the surface folding transform achieves better results than the other two transforms, namely, Cartesian and polar. Based on their experiments, we can conclude that a cancelable transform can be applied in the feature domain without much loss in performance.

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**Nalini K. Ratha** received the BTech degree in electrical engineering from the Indian Institute of Technology, Kanpur, the MTech degree in computer science and engineering also from the Indian Institute of Technology, Kanpur, and the PhD degree in computer science from Michigan State University, East Lansing. He is a research staff member at the IBM Thomas J. Watson Research Center, Hawthorne, New York, where he leads the biometrics research efforts. He has published more than 50 journal and conference papers on biometrics-related topics. He is a coauthor of the popular book on biometrics *Guide to Biometrics* and also a coeditor of the book *Automatic Fingerprint Recognition Systems*, both published by Springer. He has received several awards at IBM and has been awarded seven patents. He is coediting a upcoming special issue on biometrics in *IEEE Transactions on Systems, Man, and Cybernetics, Part B* and *IEEE Transactions on Information Forensics and Security*. He has been associated with several biometrics conferences: he was cochair of the the ICPR '06 Associated Theme: Biometrics, cochair of the IEEE Workshop on Biometrics (collocated with CVPR '06), cochair AVBPA '05, and program cochair of IEEE BTAS '07. He is a fellow of the IEEE, the IEEE Computer Society, and a member of the ACM. His research interests include biometrics recognition, pattern recognition, image processing, and architectures for computer vision systems.

**Sharat Chikkerur** received the BE degree from Bangalore University, India, and the master's degree from the State University of New York, Buffalo. He is a PhD candidate at the Massachusetts Institute of Technology. He has worked as a software engineer at Siemens Information Systems Ltd, India, and as a software engineer at Infosys, Embedded Systems and Devices Lab. His current research interests include computer vision, biometrics, and pattern recognition. He is a student member of the IEEE.

**Jonathan H. Connell** received the PhD degree in artificial intelligence from the Massachusetts Institute of Technology (MIT) in 1989, doing work on behavior-based robotics. He is a research staff member at IBM's T.J. Watson Research Center in Hawthorne, New York. He was a past associate editor for *IEEE Transactions on Pattern Analysis and Machine Intelligence*, has taught in the Cognitive Science program at Vassar College, has 13 issued patents, and has written books on robots, robot learning, and biometrics. At IBM, he has done work on mobile robot navigation, machine learning, video search, and audiovisual speech recognition. Most recently, his work has focused on computer vision for retail product verification, biometric identification, and video surveillance. He is a member of AAAI and a senior member of the IEEE.

**Ruud M. Bolle** received the bachelor's degree in analog electronics and the master's degree in electrical engineering from Delft University, Delft, The Netherlands. He received the master's degree in applied mathematics and the PhD degree in electrical engineering from Brown University, Providence, Rhode Island. Subsequently, in 1984, he joined the IBM Thomas J. Watson Center, Artifical Intelligence Department, as a research staff member. He was the manager of the Exploratory Computer Vision Group for 15 years from 1988 (the inception of the group) until 2003. Currently, his research is focused on biometrics applications such as cancelable biometrics, large-scale search systems, and performance analysis of biometric systems. In general, his research accomplishments and interests are in the areas of video processing, multimodal video databases, multimodal human-computer interaction, and biometrics and security. Dr. Bolle is a fellow of the IEEE, the IEEE Computer Society, a fellow of the International Association for Pattern Recognition, and a member of the IBM Academy of Technology. He was the recipient of the IEEE Computer Society 2000 Technical Achievement Award. Dr. Bolle coauthored the recently published book *Guide to Biometrics* (Springer-Verlag) and he served as coeditor of the books *Biometrics: Personal Identification in Networked Society* (Kluwer) and *Automatic Fingerprint Recognition Systems* (Springer-Verlag). He is an area editor of *Computer Vision and Image Understanding* and an associate editor of *Pattern Recognition*. He was guest editor of the special issue on biometrics, *IEEE Transactions on Systems, Man, and Cybernetics, Part C: Applications and Reviews*, August 2005. He has authored or coauthored more than 80 refereed technical publications.

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